1 Related Work

Research relevant to structured multi-turn interaction with large language models (LLMs) spans structured reasoning methods, dialogue modelling and memory systems, conversation disentanglement, and recent work on multi-turn interaction evaluation and training.

1.1 Structured Reasoning in LLMs

Several studies demonstrate that explicitly structured reasoning improves LLM performance. Chain-of-Thought prompting elicits intermediate reasoning steps and significantly improves accuracy on arithmetic, commonsense, and symbolic reasoning tasks Wei et al. (2022). Self-Consistency extends this approach by sampling multiple reasoning paths and aggregating outputs, improving robustness to decoding variance Wang et al. (2023). Tree-of-Thought introduces explicit search over a tree of partial solutions, enabling exploration and backtracking during inference Yao et al. (2023). Graph-of-Thought generalizes this framework by representing reasoning steps as nodes in a directed graph, allowing more flexible dependency structures Besta et al. (2023). These approaches primarily focus on single-task or single-query reasoning and do not address persistent multi-turn dialogue structure.

1.2 Dialogue Modelling and Context Management

Hierarchical dialogue models predate modern LLMs and demonstrate the benefits of multi-level structure. The hierarchical recurrent encoder–decoder (HRED) models dialogue as sequences of utterances rather than flat token streams, improving coherence over multiple turns Serban et al. (2016). However, such architectures embed structure internally and do not provide explicit mechanisms for managing topics or branches at the interaction level.

Recent work highlights fundamental limitations of long-context usage in LLMs. Empirical analysis shows that models exhibit degraded performance when relevant information appears in the middle of long input sequences, a phenomenon known as the "lost in the middle" effect Liu et al. (2023). Memory-augmented systems such as MemGPT introduce explicit memory hierarchies to manage limited context windows Packer et al. (2023), while retrieval-augmented generation (RAG) integrates external knowledge sources to improve factual grounding Lewis et al. (2020). These methods improve information access but do not directly address topic separation or conversational entanglement.

1.3 Conversation Disentanglement

Conversation disentanglement research studies the recovery of latent conversational threads from interleaved dialogue streams. Kummerfeld et al. provide benchmarks and models for disentangling multi-party chat logs, demonstrating that post hoc recovery of conversation structure is difficult and error-prone Kummerfeld and et al. (2019). This line of work addresses separation after entanglement has occurred rather than preventing entanglement during interaction.

1.4 Multi-Turn Interaction and Branching

Multi-turn interaction has recently emerged as a distinct research focus. Li et al. survey evaluation scenarios, benchmarks, and enhancement methods for multi-turn LLM interaction, identifying challenges such as conversational drift, compounding errors, and long-term coherence degradation **?**. Savage proposes Conversation Forests, a branching fine-tuning framework that generates multiple conversational continuations at each turn, demonstrating improved performance on multi-turn tasks compared to linear training schemes **?**. While this work introduces branching during training, it does not consider user-facing conversational structure.

1.5 Summary and Gap

Prior work establishes that structured reasoning, managed memory, and hierarchical modelling improve LLM performance. However, existing approaches primarily intervene at the level of prompting, inference, memory, or training. There is limited research on persistent, user-facing conversational structures that explicitly support topic isolation and parallel dialogue threads during interaction. This gap motivates hierarchical conversational architectures that structure dialogue proactively rather than relying on post hoc recovery or increased context length.

References

Besta, M. et al. (2023). Graph of thoughts: Solving elaborate problems with large language models. *arXiv preprint arXiv:2308.07998*.

Kummerfeld, J. K. and et al. (2019). Large-scale conversation disentanglement: Annotation, models, and benchmarks. In *ACL*.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., et al. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. In *NeurIPS*.

Liu, N. F. et al. (2023). Lost in the middle: How language models use long contexts. *arXiv preprint arXiv:2307.03172*.

Packer, C., Menezes, P., Raffel, C., et al. (2023). Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*.

Serban, I. V., Sordoni, A., Bengio, Y., Courville, A., and Pineau, J. (2016). Building end-to-end dialogue systems using generative hierarchical neural network models. In *AAAI*.

Wang, X., Wei, J., Schuurmans, D., Chi, E. H., Le, Q. V., and Zhou, D. (2023). Self-consistency improves chain of thought reasoning in language models. *ICLR*.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q. V., and Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*.

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T., Cao, Y., and Narasimhan, K. (2023). Tree of thoughts: Deliberate problem solving with large language models. *arXiv preprint arXiv:2305.10601*.